

Supplementary File for *In Context Learning* *Foundation Models for Materials Property* *Prediction with Small datasets*

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Supplementary Notes

S1. Pseudo-Code of ICL-FM

We present the pseudo-code of our ICL-FM algorithm for materials property prediction here. X features can be raw compositional and structural features or embedding features extracted from neural network models.

Algorithm S1 Training and Evaluation of TabPFN for Materials Property Regression

Require: Feature matrix X , target vector y , number of folds K

Ensure: Per-fold MAE and R^2 , and overall averages

- 1: Split the dataset into K folds using K-fold cross validation.
- 2: **for** $i = 1$ to K **do**
- 3: Let $(X_{\text{train}}, y_{\text{train}})$ be fold i 's training set.
- 4: Let $(X_{\text{test}}, y_{\text{test}})$ be fold i 's held-out test set.
- 5: Initialize a TabPFN regression model M with default settings.
- 6: Train TabPFN on the current fold:

$$M \leftarrow \text{Fit}(X_{\text{train}}, y_{\text{train}})$$

- 7: Predict test targets using the trained model:

$$\hat{y} = \text{Predict}(M, X_{\text{test}})$$

- 8: Compute evaluation metrics for fold i :

$$\text{MAE}_i = \text{MAE}(y_{\text{test}}, \hat{y}), \quad R_i^2 = R^2(y_{\text{test}}, \hat{y})$$

- 9: Save fold-level predictions and the trained model to disk.
 - 10: **end for**
 - 11: Compute the mean and standard deviation of MAE and R^2 across all folds.
 - 12: **return** $\{\text{MAE}_i, R_i^2\}_{i=1}^K$ and the aggregated statistics.
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S2. Baseline models for materials property prediction

A variety of specialized models have been proposed for materials property prediction, each targeting specific input modalities and learning paradigms. In the following, we briefly summarize representative models and their key contributions.

CGCNN. The Crystal Graph Convolutional Neural Network (CGCNN)¹ is a structure-based model that represents crystals as graphs of atoms connected by bonds. It applies convolutional filters over atomic neighborhoods to aggregate local structural information, effectively capturing bonding environments and enabling accurate prediction of structure-sensitive properties.

coNGN. The Composition-Normalized Graph Network (coNGN)² is a structure-based graph neural network that replaces traditional dense readout layers with a multiplicity-based aggregation function. This design allows the model to generalize more effectively across materials with varying unit cell sizes and atomic multiplicities.

DeeperGATGNN. DeeperGATGNN³ extends graph attention networks by introducing hierarchical attention mechanisms that weight both node and edge features. This architecture enhances message-passing across long-range atomic interactions and improves the representation of complex bonding patterns.

CrabNet. The Compositionally-Restricted Attention-Based Network (CrabNet)⁴ is a composition-based transformer model that employs self-attention over learned elemental embeddings. By capturing context-dependent interactions among elements, CrabNet models non-linear compositional trends without relying on structural information.

Roost. Roost⁵ is another composition-based attention model that uses message passing over compositional graphs. It learns smooth embeddings that generalize across chemical spaces, providing strong performance on small datasets and serving as a robust baseline for composition-driven property prediction.

S3. Limitations of composition-based models: The case of perovskites

Although both structure- and composition-based SOTA models perform comparably on smaller or less structurally complex datasets (see Table S2 for more details), their performance gap widens significantly as the underlying property becomes more dependent on the atomic arrangement and bonding topology. Among all benchmarks, the **perovskites** dataset provides the most striking example of this divergence.

Across the six Matbench datasets, structure-based models consistently outperform composition-only approaches, with relative performance gains ranging from 15% to more than 90%. The largest gap appears in the **perovskites** dataset, where the structure SOTA (coGN, MAE = 0.027) achieves an order-of-magnitude lower error than the best composition-only model (CrabNet, MAE = 0.407), corresponding to a 93.4% improvement.

This disparity arises because the perovskite property target (formation energy of a 5-atom perovskite cell) is strongly influenced by the detailed atomic arrangement, bonding environment, and crystal lattice distortions—information that a composition-only representation cannot capture. In other words, the same chemical composition may adopt different crystal structures or distortions with significantly different energies, so without structural descriptors, the model lacks access to the key variance in the data. As a result, structure-based models achieve dramatically better performance, whereas composition-only methods plateau at considerably higher error. The strong structure dependence observed in the **perovskites** dataset underscores the limitations of composition-only features for systems with complex lattice distortions. In contrast, some MatBench tasks exhibit the opposite trend, where the foundation model (FM) achieves the most significant performance improvement even without explicit structural inputs. Among these, the **phonons** dataset stands out as the benchmark with the highest relative gain.

Figures S1–S2 provide a more detailed examination of how individual compositional descriptors affect the ICL-FM’s predictions on the phonons dataset. Specifically, Fig. S2 shows the SHAP beeswarm plots, which visualize the direction and magnitude of per-sample feature contributions, while Fig. S3 presents feature-dependence plots that highlight pairwise interactions between the mean periodic table row and correlated features such as electronegativity.

S4. Foundation model integration with pretrained CGCNN checkpoints

To further examine the compatibility of the foundation model with pretrained graph neural networks, we conducted experiments using publicly available CGCNN checkpoints originally trained for general formation energy per atom prediction. These pretrained weights were evaluated on three MatBench tasks—shear modulus (**log_gvrh**), bulk modulus (**log_kvrrh**), and perovskite formation energy (**perovskites**). Two variants of the foundation-integrated model were tested: FMEF-P, initialized from the pretrained CGCNN checkpoint, and FMEF-B, initialized from our locally trained baseline CGCNN. Table S8 summarizes the mean absolute error (MAE) and the relative improvement (%) compared to both the MatBench benchmark and the corresponding baseline.

For the elastic property tasks, both foundation-integrated variants show consistent improvement over their respective baselines. The FMEF-P model achieves lower MAEs than the pretrained CGCNN checkpoint for **log_gvrh** (0.0929 vs. 0.1155) and **log_kvrrh** (0.0692 vs. 0.0987), demonstrating that the foundation model effectively refines pretrained graph embeddings and improves generalization for mechanically stable materials. Similarly, the FMEF-B variant outperforms the locally trained CGCNN baseline across the same tasks, indicating that foundation-based integration provides additional regularization benefits even when initialized from scratch.

In contrast, the **perovskites** dataset exhibits severe degradation for the pretrained checkpoint (MAE ≈ 2.5), which propagates to the FMEF-P model. This failure arises from a mismatch in target definition: the pretrained CGCNN was trained for the per-atom formation energy, whereas MatBench defines perovskite formation energy as the heat of formation

for a five-atom unit cell. The resulting scale and distribution shift render the pretrained features incompatible with the downstream objective. Nevertheless, the baseline-initialized FMEF-B remains numerically stable, confirming that the degradation originates from task misalignment rather than from the foundation coupling itself.

These results confirm that foundation integration consistently improves both pretrained and baseline CGCNN models when upstream and downstream targets share compatible physical definitions. This demonstrates that the foundation model can enhance pretrained graph representations through in-context reasoning, provided that the underlying property distributions remain aligned.

Supplementary Figures

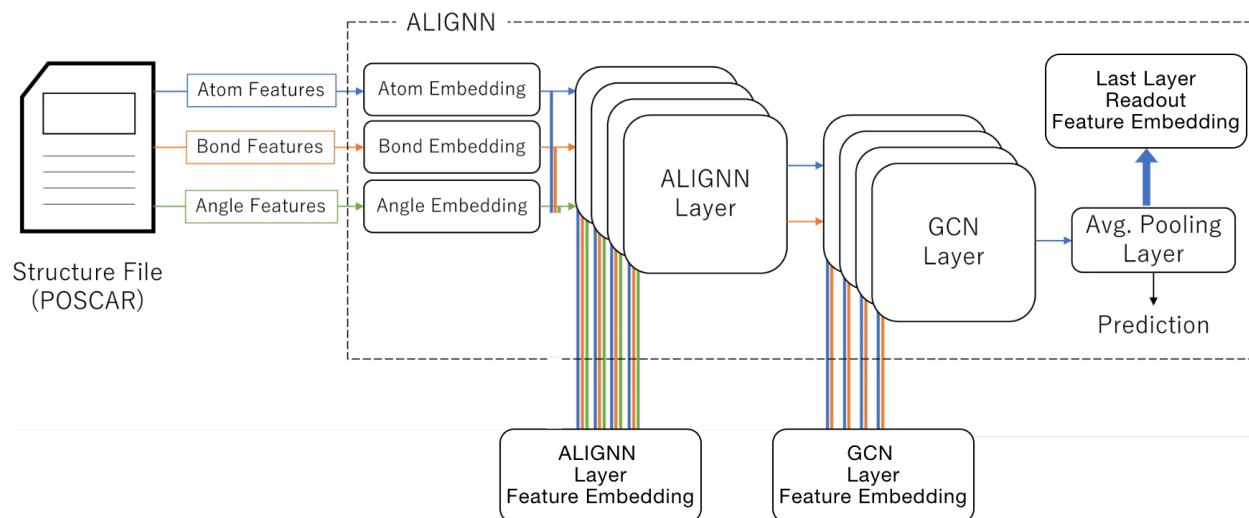
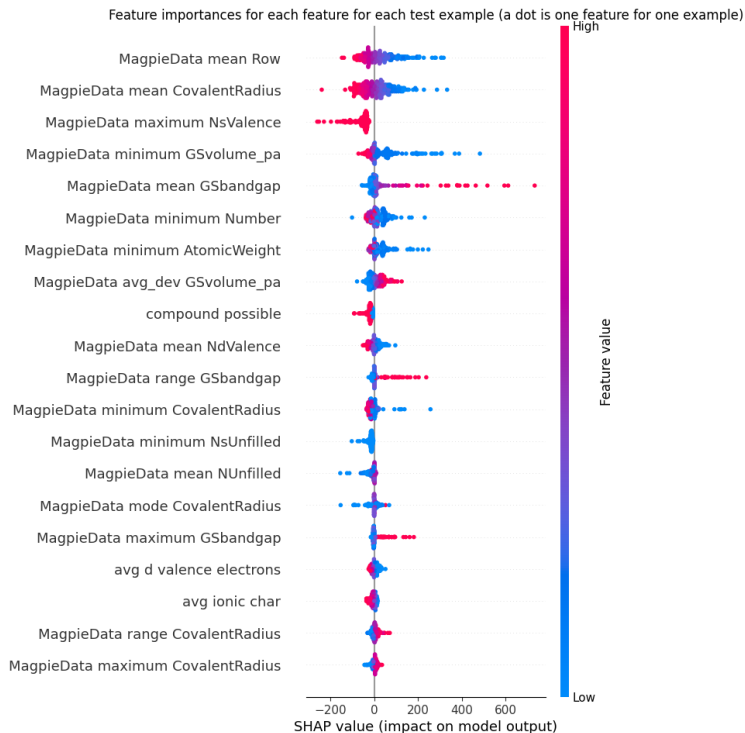
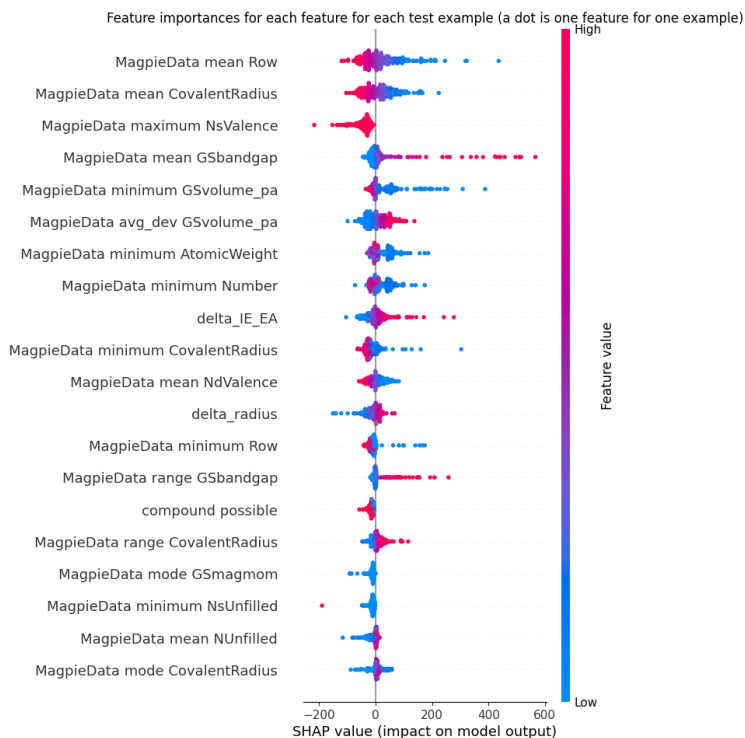


Figure S1: Schematic illustration of the ALIGNN architecture showing alternating atom, bond, and angle feature extraction layers. Each embedding (ali0-ali3, gcn0-gcn3, and readout) corresponds to a progressively deeper structural representation as summarized in Table 3. Adapted and modified from Gupta *et al.*⁶



(a) SHAP beeswarm for Magpie features.



(b) SHAP beeswarm for MagpieEX features.

Figure S2: SHAP beeswarm plots for Magpie and MagpieEX inputs on the phonons dataset. Each point represents one material instance, colored by raw feature value (red for high, blue for low). The plots reveal how variations in key descriptors modulate the predicted property, demonstrating that MagpieEX yields smoother and more physically consistent feature effects.

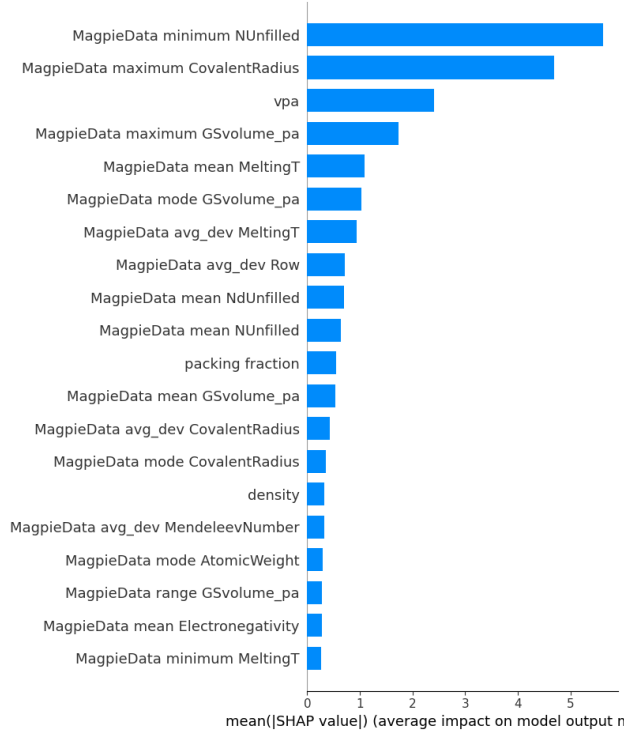


Figure S3: SHAP analysis of the Random Forest (RF) model for lattice thermal conductivity prediction. Each point represents an individual material sample, colored by the raw feature value (red for high, blue for low). Features such as NUnfilled, CovalentRadius, and vpa dominate the prediction space, confirming that the RF model relies on a small number of high-impact, axis-aligned descriptors. The concentration of feature importance at a few key variables indicates hierarchical, decision-node behavior, contrasting with the distributed representation observed in the Foundation Model (FM) SHAP analysis (Fig.4a). Together, these results highlight the complementary interpretability profiles of ensemble-based and foundation-based approaches for modeling lattice thermal conductivity.

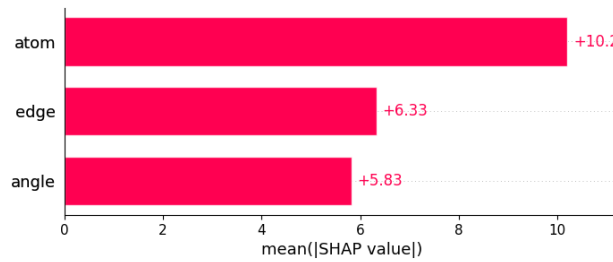


Figure S4: Mean absolute SHAP values for atom-, edge-, and angle-level features in the ALIGNN-FM data fusion model for lattice thermal conductivity prediction. The atom-level contributions dominate the total attribution (+10.2), followed by edge (+6.33) and angle (+5.83) features, indicating that the foundation model emphasizes atomic-level descriptors as primary drivers of phonon transport. The distributed importance across all three feature types highlights the FM’s ability to integrate local bonding and angular correlations into a unified, physically consistent representation of thermal conductivity.

Supplementary Tables

Table S1: Training details and hyperparameters of the ALIGNN models used in this study.

Category	Setting
Target property	Scalar regression target
Random seed	123
Data split (jdft2d example)	508 / 128 / 128
Train/validation/test ratio	0.8 / 0.1 / 0.1
Epochs	500
Batch size	32
Loss function	Mean squared error (MSE)
Optimizer	AdamW
Learning rate	1×10^{-3}
Weight decay	1×10^{-5}
Learning rate scheduler	OneCycle
Warm-up steps	2000
Neighbor strategy	k-nearest neighbors
Cutoff radius	8.0 Å
Extra cutoff	3.0 Å
Maximum neighbors	12
Line graph construction	Enabled
Model architecture	ALIGNN (atomwise)
ALIGNN layers	4
GCN layers	4
Atom input features	92 (CGCNN)
Edge input features	80
Triplet input features	40
Embedding dimension	64
Hidden dimension	256
Output dimension	1

Validation and test sets were split exclusively from the training portion of the official MatBench split to avoid data leakage.

Table S2: Comparison of SOTA models on MatBench datasets⁷ (status 2025-08-18). All results are reported as mean absolute error (MAE) over the official MatBench 5-fold splits. Ordered by ascending cardinality, these datasets are **jdft2d** (meV/atom), **phonons** (1/cm), **dielectric** (unitless), **log_gvrh** ($\log_{10}(\text{GPa})$), **log_kvrh** ($\log_{10}(\text{GPa})$), and **perovskites** (meV/unit cell).

Property	Struct. SOTA (MAE)	Compo. SOTA (MAE)	% Δ Struct.
jdft2d (meV/atom)	33.19 (MODNet)	45.61 (CrabNet)	27.2
phonons (1/cm)	28.76 (MegNet)	46.58 (Finder)	38.2
dielectric (unitless)	0.271 (MODNet)	0.320 (Finder)	15.3
log_gvrh ($\log_{10}(\text{GPa})$)	0.067 (coNGN)	0.100 (Finder)	33.0
log_kvrh ($\log_{10}(\text{GPa})$)	0.049 (coNGN)	0.076 (CrabNet)	35.5
perovskites (meV/unit cell)	0.027 (coGN)	0.407 (CrabNet)	93.4

Table S3: Comparison of CrabNet results on MatBench datasets (status 2025-08-18). All results are reported as mean absolute error (MAE). Relative improvements (%) are shown with respect to both the CrabNet baseline and the MatBench model.

Property	matbench	baseline	FM (CrabNet)	% Δ baseline	% Δ SOTA
jdft2d (meV/atom)	45.6104	46.5696	45.0561	3.25	1.22
phonons (1/cm)	55.1114	65.9339	63.0893	4.31	-35.46
dielectric (unitless)	0.3234	0.3479	0.3465	0.43	-8.13
log_gvrh ($\log_{10}(\text{GPa})$)	0.1014	0.1040	0.1053	-1.26	-5.76
log_kvrh ($\log_{10}(\text{GPa})$)	0.0758	0.0763	0.0804	-5.37	-6.05
perovskites (meV/unit cell)	0.4065	0.4093	0.4217	-3.02	-3.74

Table S4: Comparison of Roost results on MatBench datasets (status 2025-08-18). All results are reported as mean absolute error (MAE). Relative improvements (%) are shown with respect to both the Roost baseline and the MatBench model.

Property	Data Size	matbench SOTA	Roost baseline	FM+Roost	% Δ Roost baseline	% Δ SOTA
jdft2d (meV/atom)	636	45.6104	49.6084	49.0792	1.07	-7.61
phonons (1/cm)	1,265	46.5751	63.2299	57.9518	8.35	-24.43
dielectric (unitless)	4,764	0.3204	0.3561	0.3526	1.00	-10.03
log_gvrh ($\log_{10}(\text{GPa})$)	10,987	0.0996	0.1091	0.1077	1.22	-8.17
log_kvrh ($\log_{10}(\text{GPa})$)	10,987	0.0758	0.0824	0.0836	-0.01	-10.30
perovskites (meV/unit cell)	18,928	0.4065	0.6242			

Table S5: Comparison of CGCNN results on MatBench datasets (status 2025-08-18). All results are reported as mean absolute error (MAE). Relative improvements (%) are shown with respect to both the baseline and the MatBench model.

Property	baseline	matbench	FM+Baseline	% Δ baseline	% Δ matbench
jdft2d (meV/atom)	49.2440	58.0545	51.4114	11.44	-4.40
phonons (1/cm)	57.7635	127.2627	85.3367	32.94	-47.73
dielectric (unitless)	0.5988	0.4959	0.3554	28.34	40.65
log_gvrh ($\log_{10}$ (GPa))	0.0895	0.1006	0.0992	1.40	-10.86
log_kvrh ($\log_{10}$ (GPa))	0.0712	0.0830	0.0788	5.01	-10.70
perovskites (meV/unit cell)	0.0452	0.0965	0.0962	0.32	-112.77

Table S6: Comparison of coNGN results on MatBench datasets (status 2025-08-18). All results are reported as mean absolute error (MAE). Relative improvements (%) are shown with respect to both the baseline and the MatBench model.

Property	baseline	matbench	FM+Baseline	% Δ baseline	% Δ matbench
jdft2d (meV/atom)	36.1698	40.7246	41.0166	-0.72	-13.40
phonons (1/cm)	28.8874	37.2981	46.8690	-25.66	-62.25
dielectric (unitless)	0.3142	0.2853	0.3066	-7.47	2.42
log_gvrh ($\log_{10}$ (GPa))	0.0670	0.0711	0.0791	-11.31	-18.06
log_kvrh ($\log_{10}$ (GPa))	0.0491	0.0504	0.0579	-15.00	-17.99
perovskites (meV/unit cell)	0.0290	0.0305	0.0377	-23.61	-29.84

Table S7: Comparison of DeeperGATGNN results on MatBench datasets (status 2025-08-18). All results are reported as mean absolute error (MAE). Relative improvements (%) are shown with respect to both the baseline and the MatBench model.

Property	baseline	matbench	FM+Baseline	% Δ baseline	% Δ matbench
jdft2d (meV/atom)	45.4611	48.1363	45.0457	6.42	0.91
phonons (1/cm)	33.1211	44.5217	45.1159	-1.33	-36.22
dielectric (unitless)	0.3355	0.3635	0.3454	4.97	-2.95
log_gvrh ($\log_{10}$ (GPa))	0.0903	0.0962	0.0943	1.97	-4.45
log_kvrh ($\log_{10}$ (GPa))	0.0698	0.0768	0.0691	10.03	1.05
perovskites (meV/unit cell)	0.0288	0.0472	0.0443	6.07	-53.91

Table S8: Comparison of CGCNN pretrained and foundation-integrated models on selected MatBench tasks (status 2025-08-18). The pretrained CGCNN checkpoint was trained on a general **formation energy per atom** target, which differs from the MatBench **perovskites** definition (heat of formation for the entire 5-atom perovskite cell). All results are reported as mean absolute error (MAE). Relative improvements (%) are computed with respect to both the MatBench benchmark and the corresponding baseline.

Property	MatBench	Pretrain*	FMEF-P	% Δ MatBench	% Δ Baseline	Baseline	FMEF-B	% Δ Baseline	% Δ MatBench
Shear modulus ($\log_{10}$ (GPa))	0.0895	0.1155	0.0929	19.59	-3.78	0.1006	0.0992	1.40	-10.86
Bulk modulus ($\log_{10}$ (GPa))	0.0712	0.0987	0.0692	29.85	2.75	0.0830	0.0788	5.01	-10.70
Perovskites (meV/unit cell)	0.0452	2.5365	0.1258	95.04	-178.35	0.0965	0.0962	0.32	-112.77

Table S9: Cation–anion contrast and bonding-related descriptors introduced in MagpieEX. All features are computed from elemental properties and oxidation-state assignments as implemented in the published codebase.

#	Feature name	Description / definition	Units
–	Summary counts	Standard Magpie feature set: 145 features (baseline). MagpieEX adds 15 cation–anion contrast descriptors (this table), for a total of 160 features.	–
1	<code>delta_en</code>	Difference between mean anion and mean cation electronegativity: $\langle\chi_{\text{anion}}\rangle - \langle\chi_{\text{cation}}\rangle$.	unitless
2	<code>delta_radius</code>	Difference between mean anion and mean cation atomic radius: $\langle r_{\text{anion}}\rangle - \langle r_{\text{cation}}\rangle$.	Å
3	<code>delta_valence_s</code>	Difference in mean <i>s</i> -orbital valence occupancy between cations and anions.	unitless
4	<code>delta_valence_p</code>	Difference in mean <i>p</i> -orbital valence occupancy between cations and anions.	unitless
5	<code>delta_valence_d</code>	Difference in mean <i>d</i> -orbital valence occupancy between cations and anions.	unitless
6	<code>delta_valence_f</code>	Difference in mean <i>f</i> -orbital valence occupancy between cations and anions.	unitless
7	<code>delta_IE_EA</code>	Difference between mean cation ionization energy and mean anion electron affinity.	eV
8	<code>delta_polarizability</code>	Difference between mean anion and mean cation atomic polarizability.	atomic units
9	<code>bond_ionicity</code>	Ionicity proxy derived from electronegativity contrast, $1 - \exp(-0.25 \cdot \Delta\chi^2)$.	unitless
10	<code>avg_cation_oxi</code>	Average oxidation state of cation species.	unitless
11	<code>avg_anion_oxi</code>	Average oxidation state of anion species.	unitless
12	<code>oxi_diff</code>	Difference between average cation and anion oxidation states.	unitless
13	<code>cation_lonepair_flag</code>	Binary indicator for the presence of cations with stereochemically active lone pairs (e.g., Pb, Bi, Sn, Tl).	0/1
14	<code>anion_lonepair_flag</code>	Binary indicator for the presence of anions with stereochemically active lone pairs (e.g., Se, Te, I).	0/1
15	<code>cation_anion_size_ratio</code>	Ratio of mean anion radius to mean cation radius, $\langle r_{\text{anion}}\rangle / \langle r_{\text{cation}}\rangle$.	unitless

Notes: Cation and anion species are identified based on oxidation-state assignments obtained from pymatgen. All group averages are computed over species within each category. Descriptors that cannot be computed due to missing oxidation-state assignments or elemental properties are assigned NaN values. The complete implementation, including oxidation-state handling and feature computation, is available in the publicly released code repository.

Table S10: Global structural descriptors introduced in the Magpie+Struct1 representation. These features are appended to the composition-only Magpie baseline and are computed directly from the crystal structure.

#	Feature name	Description / definition	Units
–	Summary counts	Magpie+Struct1 adds 7 structure-derived descriptors to the 145-dimensional Magpie composition baseline, resulting in a total of 152 features .	–
1	density	Crystal mass density computed from atomic masses and unit-cell volume.	g/cm ³
2	volume_per_atom	Unit-cell volume divided by the number of atoms in the primitive cell.	Å ³ /atom
3	packing_fraction	Fraction of the unit-cell volume occupied by atomic spheres, serving as a coarse measure of atomic packing efficiency.	unitless
4	spacegroup_number	International space group number encoding crystallographic symmetry.	unitless
5	num_symmetry_ops	Number of symmetry operations associated with the space group.	unitless
6	crystal_system_numeric	Numeric encoding of the crystal system (e.g., cubic, tetragonal, orthorhombic) as returned by <code>GlobalSymmetryFeatures</code> .	unitless
7	point_group_numeric	Numeric encoding of the crystallographic point group symmetry.	unitless

Notes: All symmetry-related descriptors are obtained from `matminer`’s `GlobalSymmetryFeatures` featurizer, with categorical or string-valued outputs removed. Only numeric symmetry attributes are retained, consistent with the implementation in the published code. Descriptors that cannot be computed for a given structure are assigned NaN values and handled during preprocessing.

Table S11: Additional local and topological structural descriptors introduced in the Magpie+Struct2 representation. These features extend Magpie+Struct1 by incorporating measures of structural heterogeneity, interatomic distance statistics, and radial distribution functions.

#	Feature group	Description / definition	Units
-	Summary counts	Magpie+Struct2 adds 117 additional structure-derived descriptors on top of Magpie+Struct1, resulting in a total of 269 features .	-
1	Structural heterogeneity	Statistical measures capturing variability in local coordination environments and bond-length distributions across atomic sites, as computed by the StructuralHeterogeneity featurizer.	unitless
2	Minimum relative distances	Flattened statistics of minimum relative interatomic distances for distinct atomic pairs, capturing short-range packing constraints and nearest-neighbor separation patterns, computed using the MinimumRelativeDistances featurizer and sanitized into numeric form.	Å
3	Radial distribution function (RDF)	Histogram-based radial distribution function features computed up to a cutoff of 10 Å with a bin size of 0.1 Å, yielding 100 bins that encode medium-range atomic ordering and coordination-shell structure.	unitless (histogram bins)

Notes: Magpie+Struct2 includes all features from Magpie+Struct1 and further augments them with local and medium-range structural descriptors. The 117 added features consist of 9 structural heterogeneity descriptors, 8 minimum-relative-distance descriptors, and 100 RDF bins, consistent with the published implementation. Nested or non-numeric outputs from structural featurizers are flattened and sanitized following the utility functions provided in the codebase. Descriptors that cannot be computed for a given structure are assigned numeric placeholders and handled consistently during preprocessing.

Table S12: Architecture and training hyperparameters of the multilayer perceptron (MLP) baseline used for comparison.

Component	Specification
Implementation	<code>sklearn.neural_network.MLPRegressor</code>
Hidden layers	(256, 128, 64)
Activation function	ReLU
Output layer	Linear (single neuron)
Optimizer	Adam
Initial learning rate	1×10^{-3}
Maximum iterations	1000
Loss function	Mean squared error (MSE)
Random seed	42
Other hyperparameters	Default <code>MLPRegressor</code> settings

Table S13: Selected leaderboard results for composition-only MatBench tasks, showing only top-performing models for each benchmark. All values are reported with four decimal places. **ICL-FM** denotes the proposed in-context learning foundation model; Underlined entries indicate results obtained using ICL-FM. Metrics follow the standard MatBench evaluation protocol for each task.

matbench_steels — regression (312 samples)				
Algorithm	mean mae	std mae	mean rmse	max_error
TPOT-Mat	79.9468	13.5883	115.3675	522.6191
AutoML-Mat	82.3043	8.8565	114.0577	463.0130
ICL-FM	<u>83.6048</u>	<u>10.4426</u>	<u>136.1758</u>	<u>1083.7610</u>
MODNet (v0.1.12)	87.7627	12.2188	144.7722	1121.0504
matbench_expt_gap — regression (4,604 samples)				
Algorithm	mean mae	std mae	mean rmse	max_error
Darwin	0.2865	0.0083	0.7185	7.6000
ICL-FM	<u>0.3142</u>	<u>0.0165</u>	<u>0.6916</u>	<u>7.2413</u>
Ax/SAASBO CrabNet v1.2.7	0.3310	0.0071	0.8123	11.1001
matbench_expt_is_metal — classification (4,921 samples)				
Algorithm	mean rocauc	std rocauc	mean f1	mean balanced_accuracy
Darwin	0.9598	0.0041	0.9599	0.9598
ICL-FM	<u>0.9366</u>	<u>0.0065</u>	<u>0.9361</u>	<u>0.9366</u>
AMMExpress v2020	0.9209	0.0028	0.9200	0.9209
matbench_glass — classification (5,680 samples)				
Algorithm	mean rocauc	std rocauc	mean f1	mean balanced_accuracy
MODNet (v0.1.12)	0.9603	0.0075	0.9784	0.9603
AMMExpress v2020	0.8607	0.0199	0.9043	0.8607
RF-SCM/Magpie	0.8587	0.0158	0.9278	0.8587
ICL-FM	<u>0.8577</u>	<u>0.0135</u>	<u>0.9276</u>	<u>0.8577</u>
MODNet (v0.1.10)	0.8107	0.0212	0.9104	0.8107

Note: All results in this table are based on the standard Magpie composition descriptors, which are the default input representation for composition-only MatBench tasks.

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